Audio Visualiser DSP Board

https://github.com/drewsum/audio_visualiser

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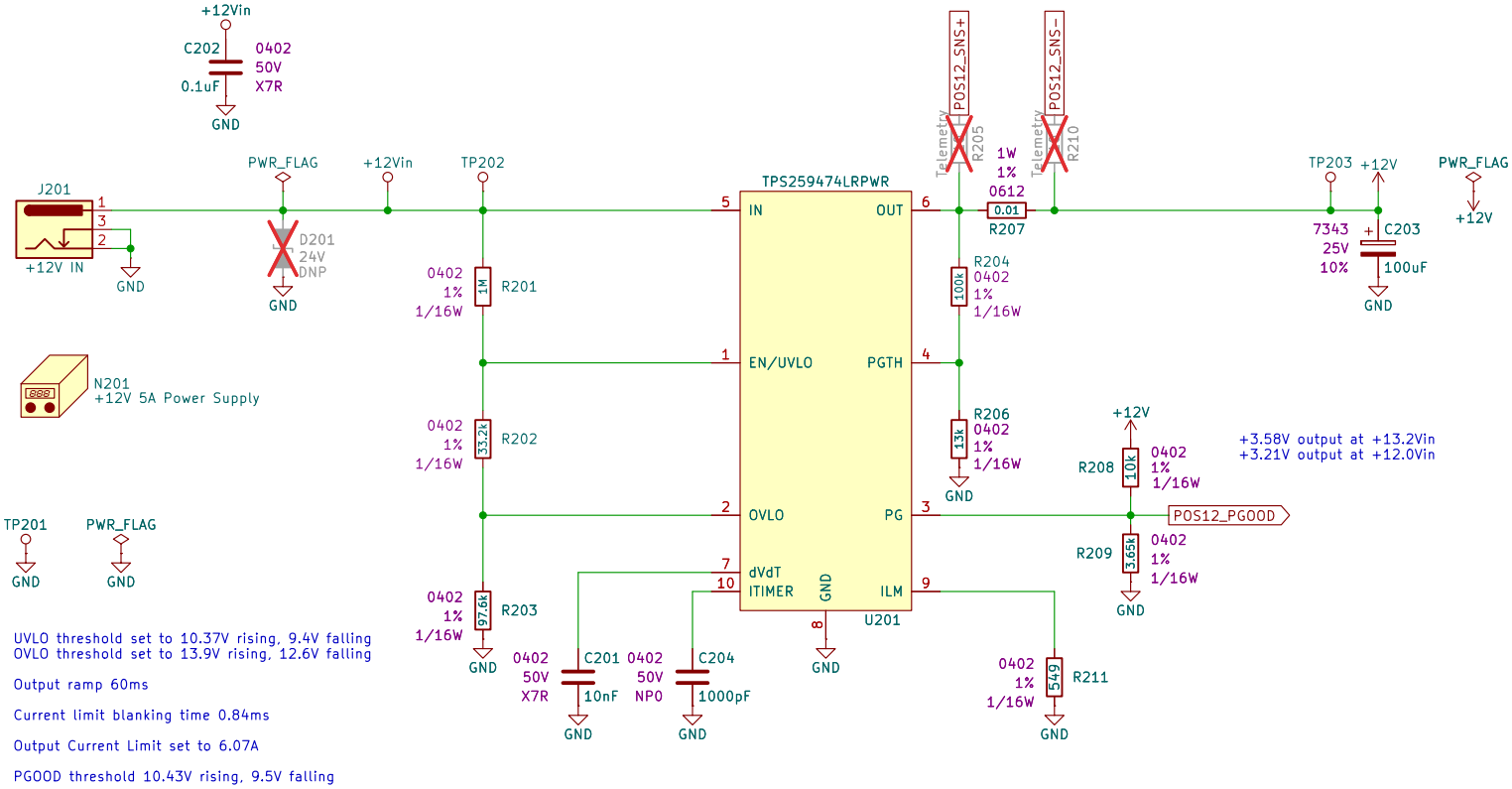
21. LED Panel Connectors — LED Panel Connectors — File: led_panel_connectors.kicad_sch

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Sheet: /
File: Audio_Visualiser_DSP.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	Rev: A
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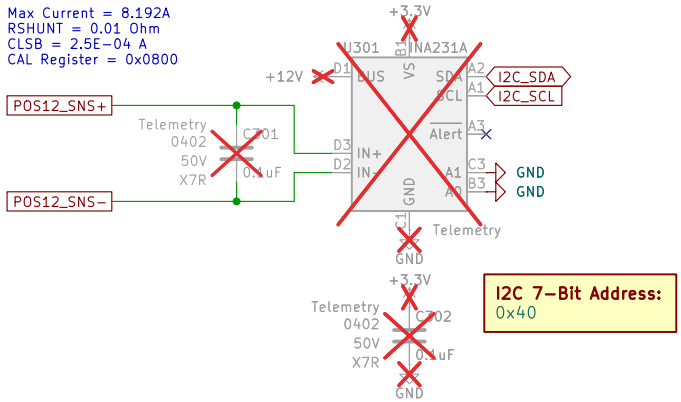
Load	Draw	Quantity	Total
LED Panel VLED 256 * 0.02A 8 3V = 15.36W / 12V = 1.28A / 80% efficiency	1.6A	2	3.2A
LED Panel POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	2	0.688A
Local POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	1	0.344A
Total		50.784W	4.232A

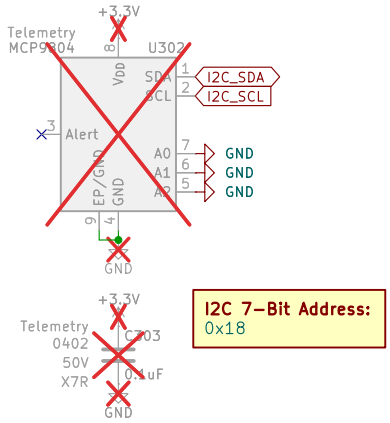
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Sheet: /12V Power Input/
File: POS12_power_input.kicad_sch

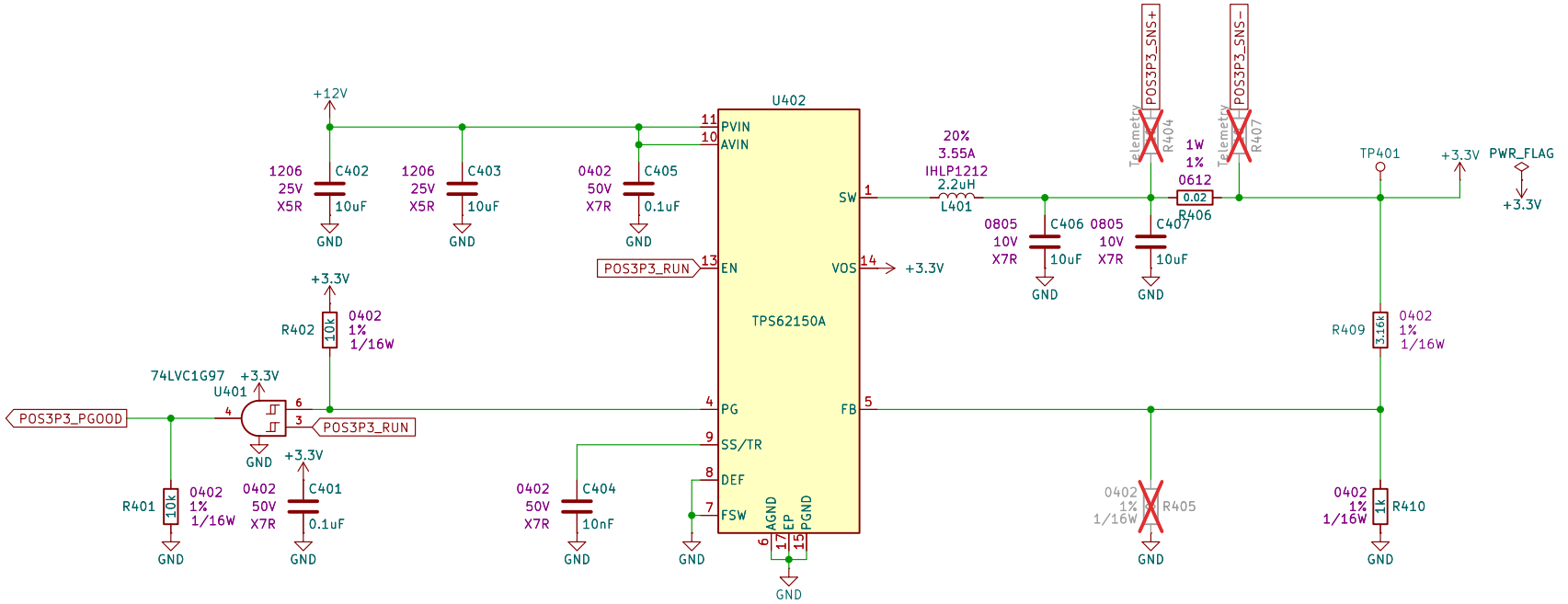
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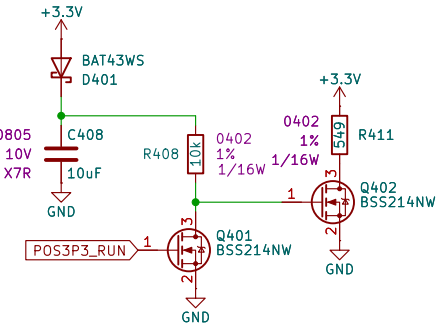
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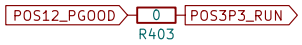




Drew Maatman and Drew Koetting		
Sheet: /12V Telemetry/		
File: POS12_telemetry.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
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Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	6	6.6mA
MCP6024	5.4mA	3	16.2mA
Analog Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V Reference	0.2mA	1	0.2mA
PCA9536 GPIO Expander	0.29mA	1	0.29mA
I2C Buffers	8.3mA	3	24.9mA
Octal Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary Encoder Module	50mA	1	50mA
LEDs	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap Touch Sensor	0.729mA	1	0.729mA
Total		0.579W	175.48mA

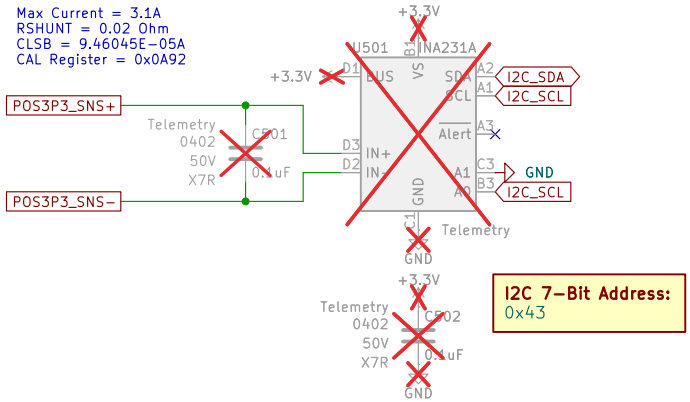
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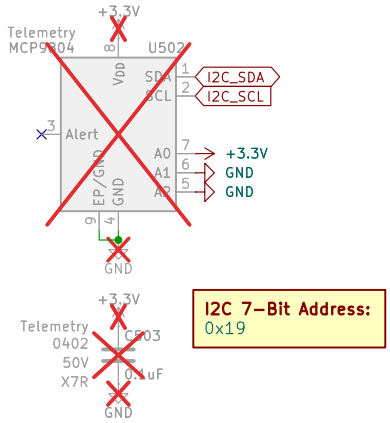
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File: POS3P3_power_supply.kicad_sch

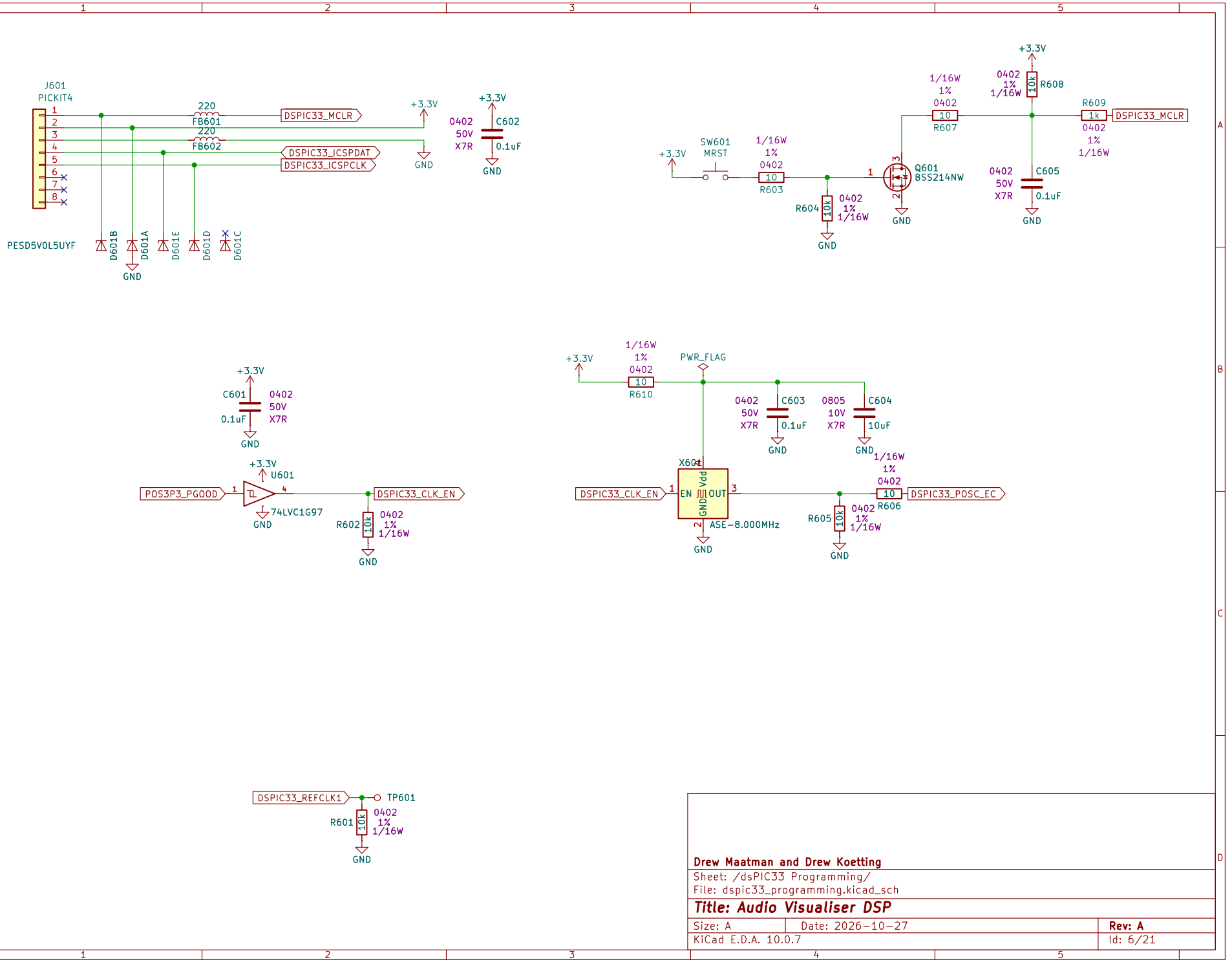
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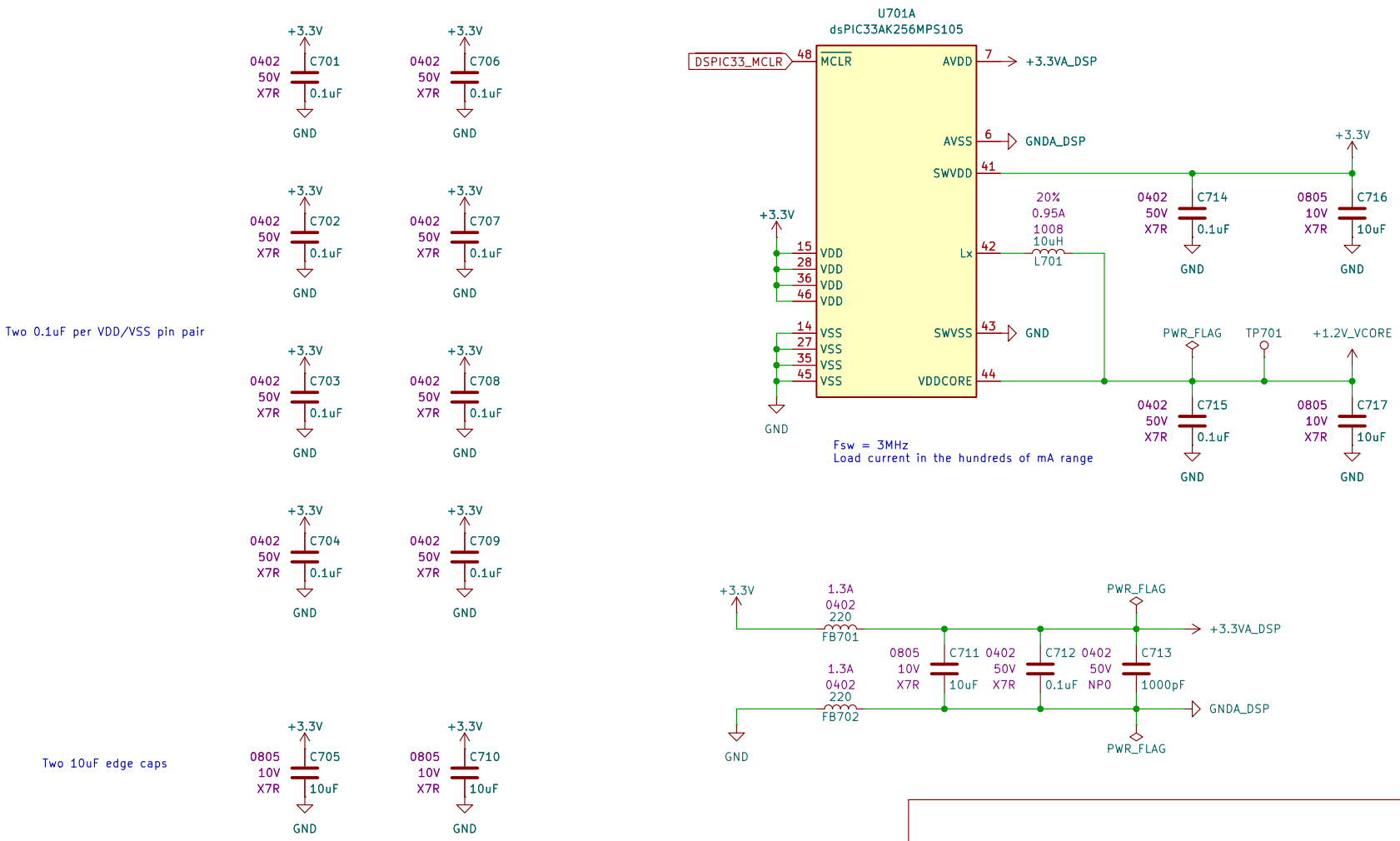
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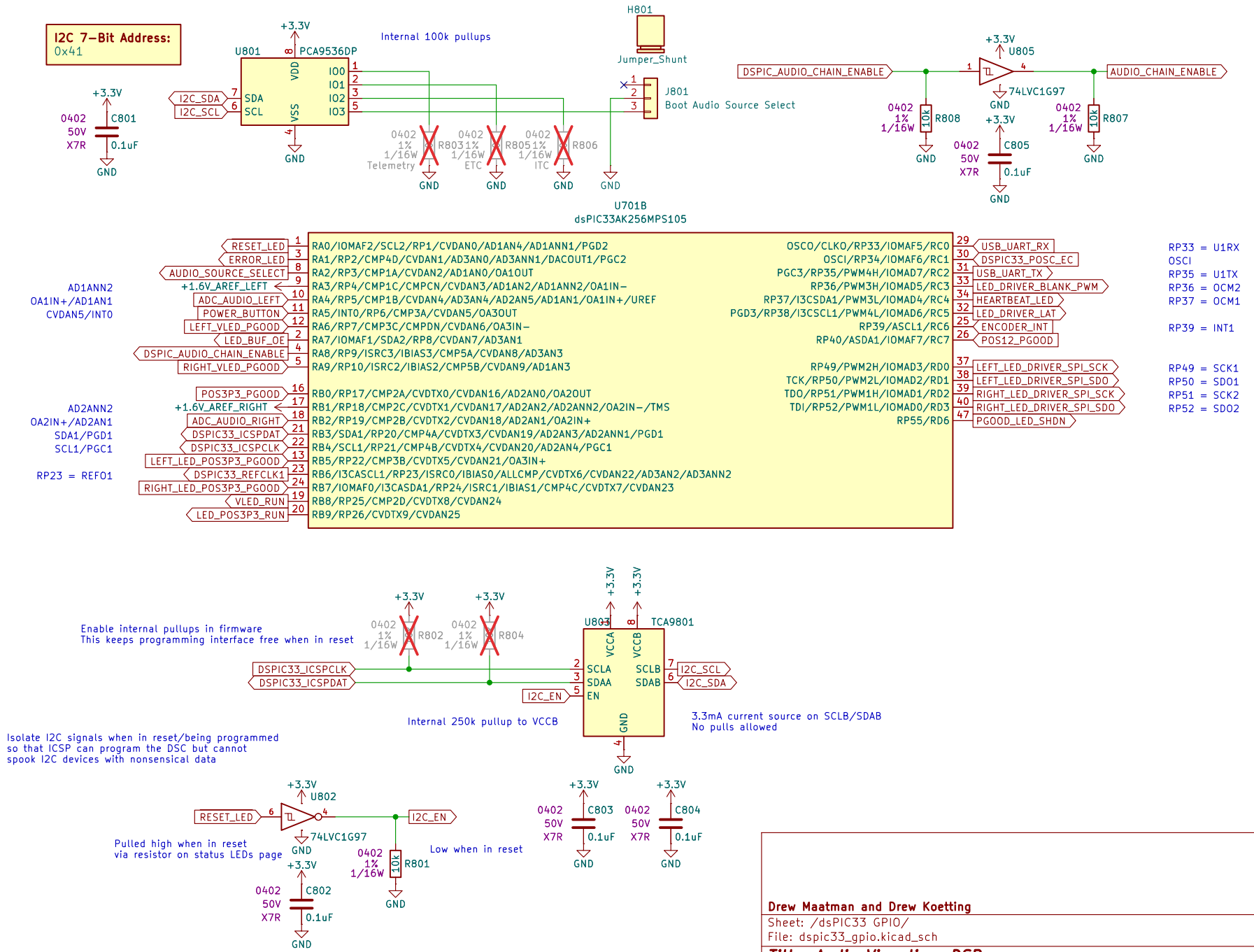


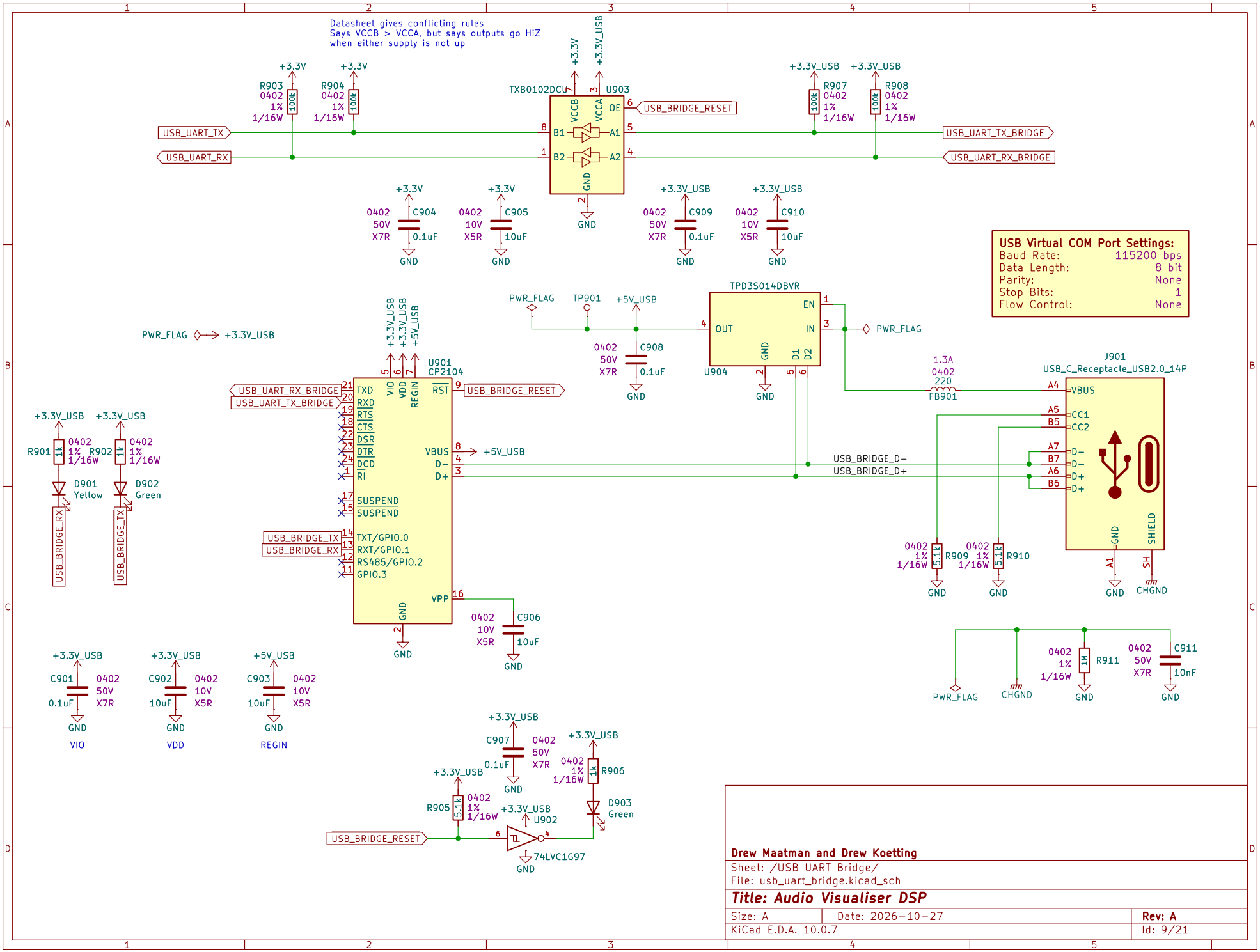
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KiCad E.D.A. 10.0.7		Id: 5/21

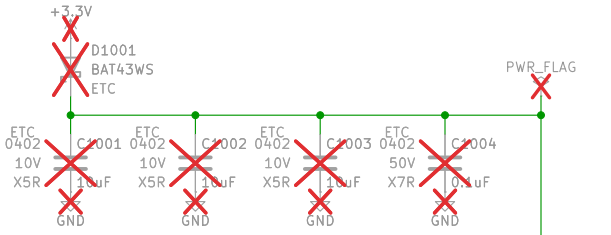

J601
PICKIT4
1
2
3
4
5
6
7
8
220
FB601
220
FB602
DSPIC33_MCLR
DSPIC33_ICSPDAT
DSPIC33_ICSPCLK
+3.3V
GND
+3.3V
0402
50V
X7R
C602
0.1uF
GND
PESD5V0L5UYF
D601B
D601A
D601E
D601D
D601C
GND
+3.3V
1/16W
1%
0402
10
R607
0402
1%
1/16W
10k
R608
R609
1k
DSPIC33_MCLR
0402
1%
1/16W
+3.3V
SW601
MRST
1/16W
1%
0402
10
R603
Q601
BSS214NW
R604
10k
0402
1%
1/16W
GND
0402
50V
X7R
C605
0.1uF
GND
+3.3V
C601
0402
50V
X7R
0.1uF
GND
+3.3V
U601
POS3P3_PGOOD
1
TL
4
DSPIC33_CLK_EN
74LVC1G97
GND
R602
10k
0402
1%
1/16W
GND
+3.3V
1/16W
1%
0402
10
R610
PWR_FLAG
0402
50V
X7R
C603
0.1uF
GND
0805
10V
X7R
C604
10uF
GND
X601
DSPIC33_CLK_EN
1
EN
Vdd
OUT
3
GND
2
ASE-8.000MHz
GND
1/16W
1%
0402
10
R606
DSPIC33_POSC_EC
R605
10k
0402
1%
1/16W
GND
DSPIC33_REFCLK1
TP601
R601
10k
0402
1%
1/16W
GND
Drew Maatman and Drew Koetting
Sheet: /dsPIC33 Programming/
File: dspic33_programming.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
Rev: A
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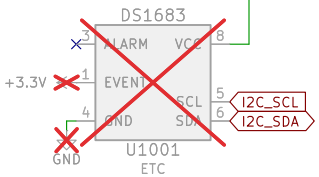


Drew Maatman and Drew Koetting		
Sheet: /dsPIC33 Power/		
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Size: A	Date: 2026-10-27	**Rev: A**
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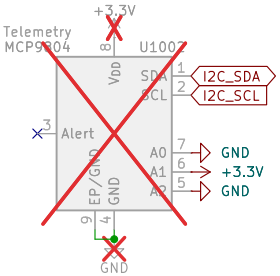

I2C 7-Bit Address:
0x41
U801
PCA9536DP
Internal 100k pullups
H801
Jumper_Shunt
J801
Boot Audio Source Select
Telemetry
ETC
ITC
U805
74LVC1G97
DSPIC_AUDIO_CHAIN_ENABLE
AUDIO_CHAIN_ENABLE
U701B
dsPIC33AK256MPS105
RP33 = U1RX
RP35 = U1TX
RP36 = OCM2
RP37 = OCM1
RP39 = INT1
RP49 = SCK1
RP50 = SDO1
RP51 = SCK2
RP52 = SDO2
RP23 = REFO1
Enable internal pullups in firmware
This keeps programming interface free when in reset
U803
TCA9801
Internal 250k pullup to VCCB
3.3mA current source on SCLB/SDAB
No pulls allowed
Isolate I2C signals when in reset/being programmed
so that ICSP can program the DSC but cannot
spook I2C devices with nonsensical data
U802
Pulled high when in reset
via resistor on status LEDs page
Low when in reset
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Sheet: /dsPIC33 GPIO/
File: dspic33_gpio.kicad_sch
Title: Audio Visualiser DSP
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Datasheet gives conflicting rules
Says VCCB > VCCA, but says outputs go HiZ
when either supply is not up
TXB0102DCUR
U903
USB_BRIDGE_RESET
USB_UART_TX
USB_UART_RX
USB_UART_TX_BRIDGE
USB_UART_RX_BRIDGE
R903
R904
R907
R908
C904
C905
C909
C910
USB Virtual COM Port Settings:
Baud Rate: 115200 bps
Data Length: 8 bit
Parity: None
Stop Bits: 1
Flow Control: None
TPD3S014DBVR
U904
PWR_FLAG
TP901
C908
U901
CP2104
USB_UART_RX_BRIDGE
USB_UART_TX_BRIDGE
USB_BRIDGE_RESET
USB_BRIDGE_D−
USB_BRIDGE_D+
J901
USB_C_Receptacle_USB2.0_14P
FB901
R909
R910
R901
R902
D901
Yellow
D902
Green
USB_BRIDGE_RX
USB_BRIDGE_TX
C901
C902
C903
C906
C911
R911
R905
R906
D903
Green
U902
74LVC1G97
VIO
VDD
REGIN
CHGND
Drew Maatman and Drew Koetting
Sheet: /USB UART Bridge/
File: usb_uart_bridge.kicad_sch
Title: Audio Visualiser DSP
Size: A
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I2C 7-Bit Address:
0x6B



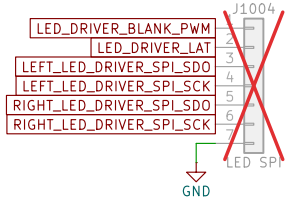


I2C 7-Bit Address:
0x1A

Ambient, place away from power components

Test Points



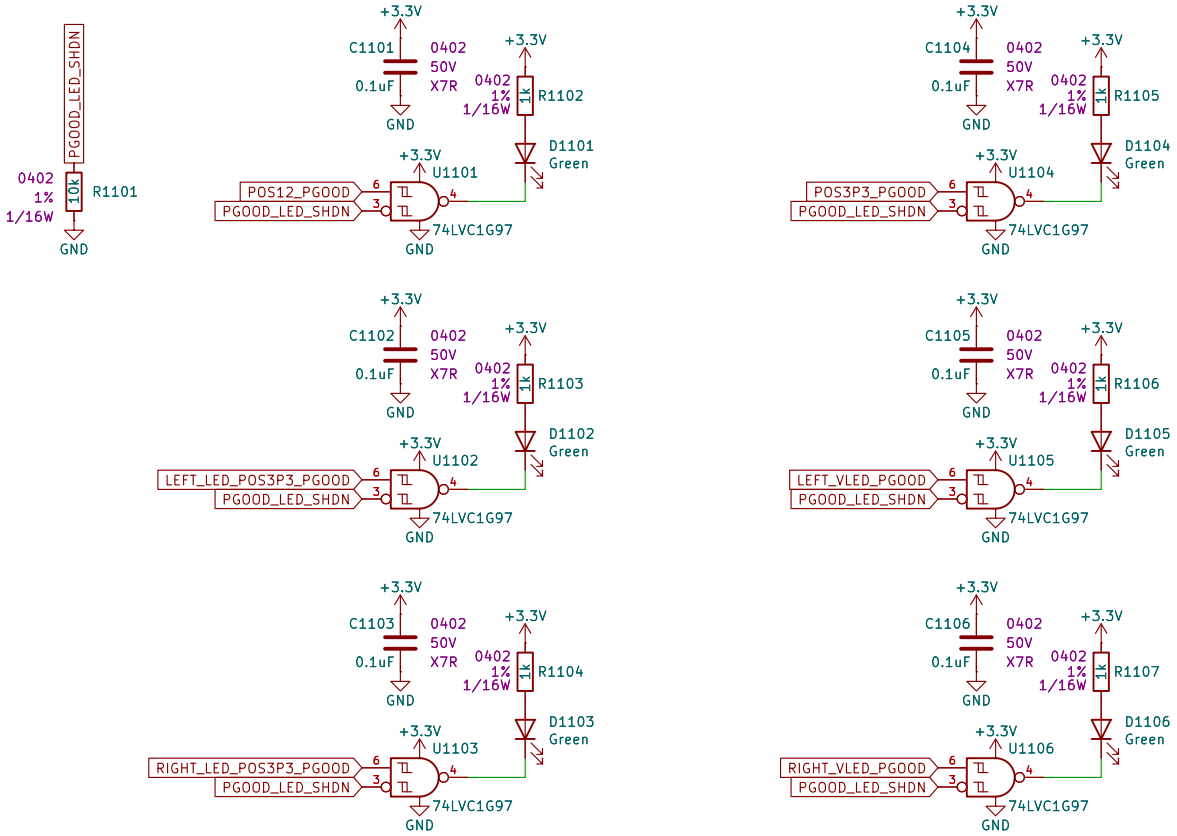


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Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
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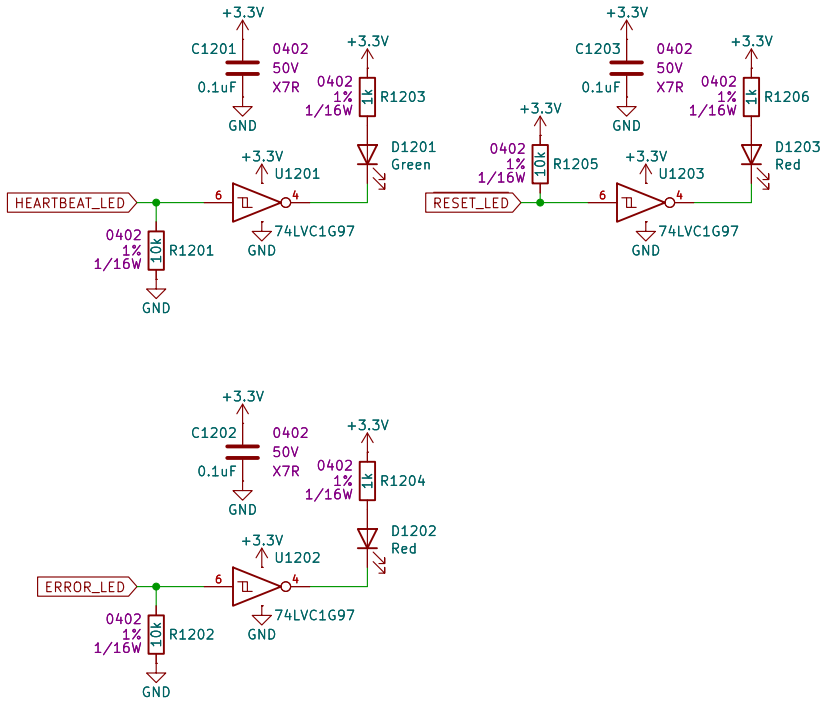


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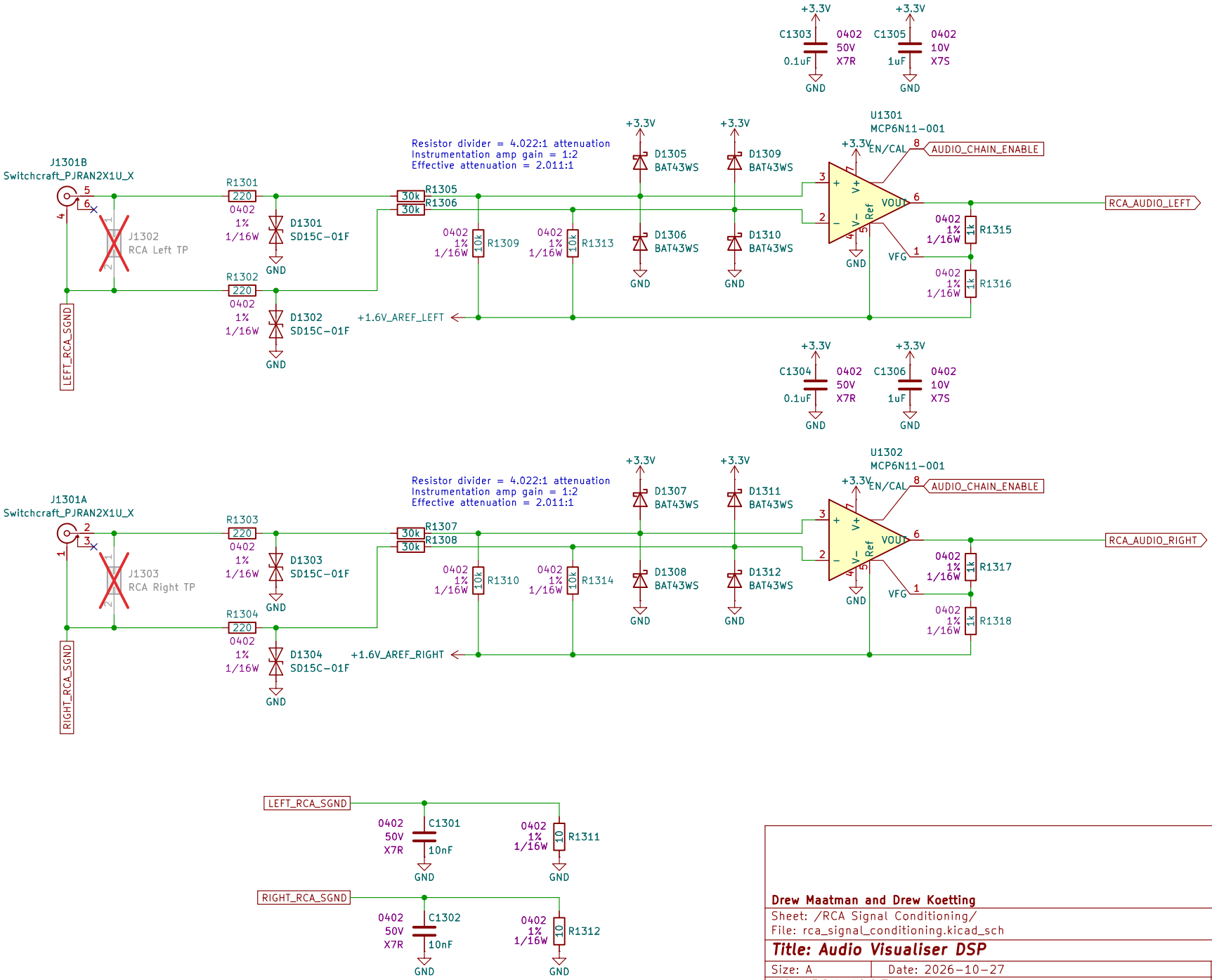
Sheet: /PGOOD LEDs/
File: pgood_leds.kicad_sch

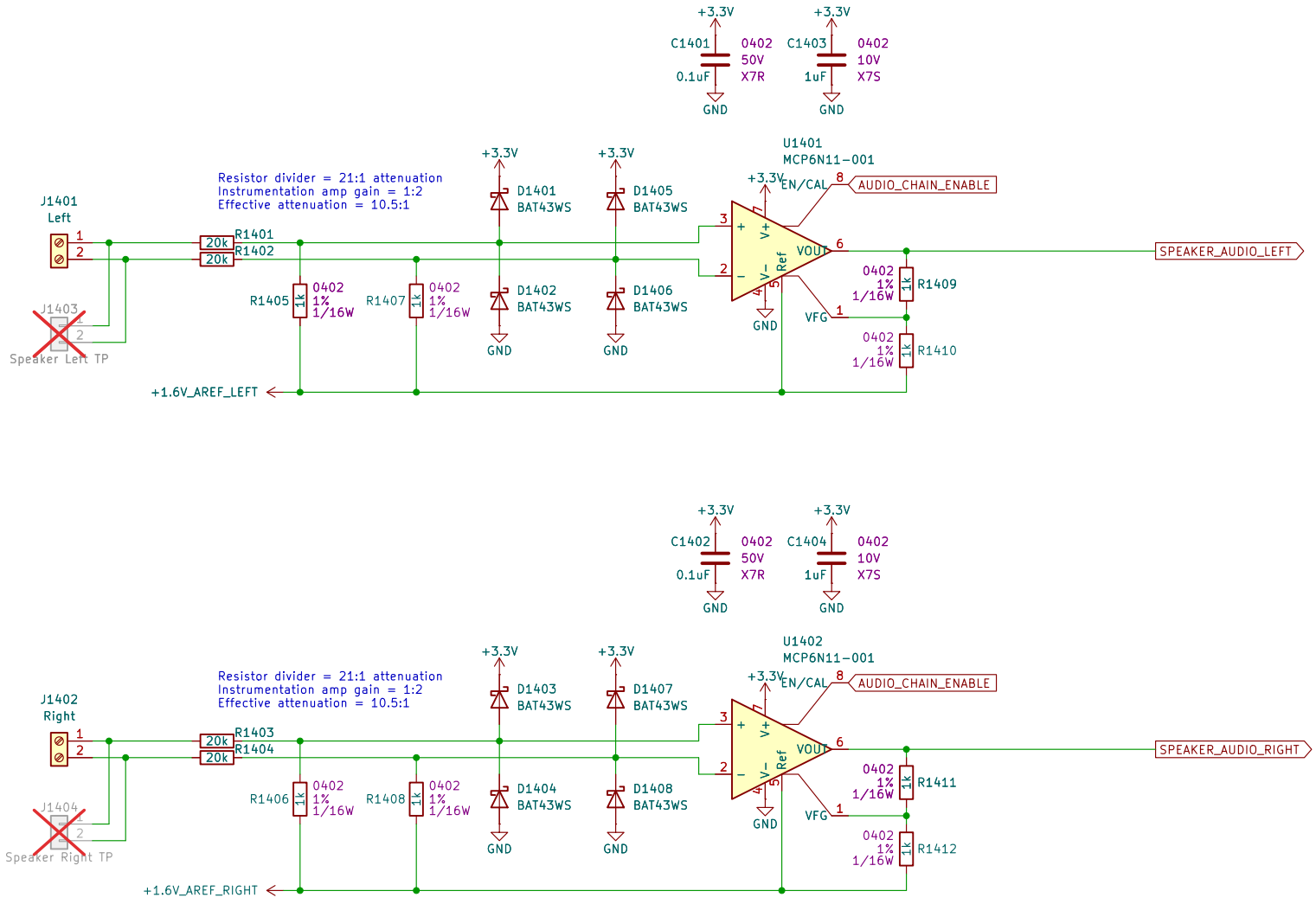
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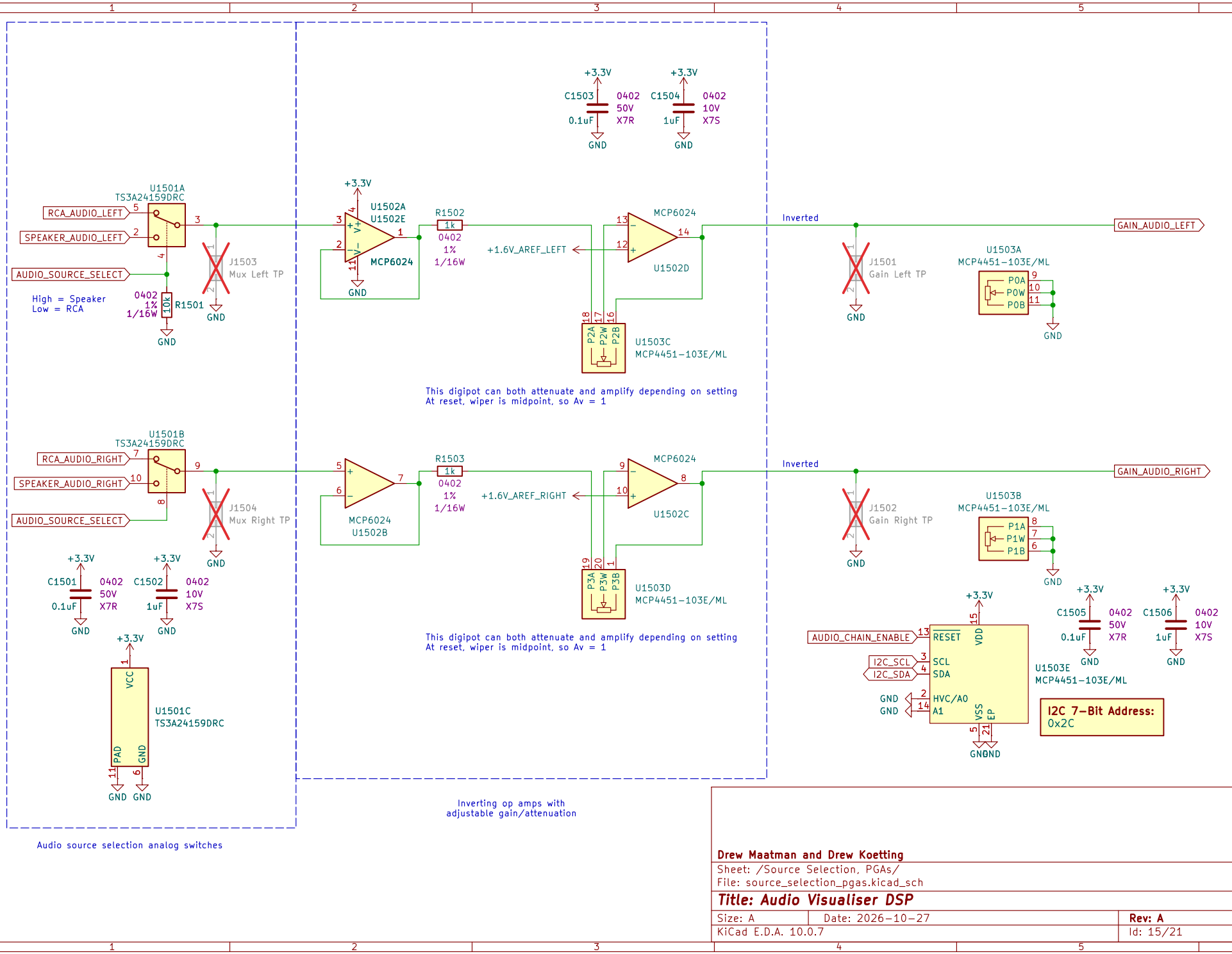


Drew Maatman and Drew Koetting		
Sheet: /Status LEDs/		
File: status_leds.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
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+3.3V
C1303
0402
50V
X7R
0.1uF
GND
+3.3V
C1305
0402
10V
X7S
1uF
GND
U1301
MCP6N11-001
EN/CAL
AUDIO_CHAIN_ENABLE
VOUT
VFG
V+
V-
Ref
RCA_AUDIO_LEFT
J1301B
Switchcraft_PJRAN2X1U_X
J1302
RCA Left TP
LEFT_RCA_SGND
R1301
220
0402
1%
1/16W
R1302
D1301
SD15C-01F
D1302
Resistor divider = 4.022:1 attenuation
Instrumentation amp gain = 1:2
Effective attenuation = 2.011:1
R1305
30k
R1306
R1309
10k
R1313
D1305
BAT43WS
D1306
D1309
D1310
R1315
1k
R1316
+1.6V_AREF_LEFT
C1304
C1306
U1302
MCP6N11-001
RCA_AUDIO_RIGHT
J1301A
Switchcraft_PJRAN2X1U_X
J1303
RCA Right TP
RIGHT_RCA_SGND
R1303
R1304
D1303
D1304
R1307
R1308
R1310
R1314
D1307
D1308
D1311
D1312
R1317
R1318
+1.6V_AREF_RIGHT
LEFT_RCA_SGND
C1301
10nF
R1311
10
RIGHT_RCA_SGND
C1302
R1312
Drew Maatman and Drew Koetting
Sheet: /RCA Signal Conditioning/
File: rca_signal_conditioning.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
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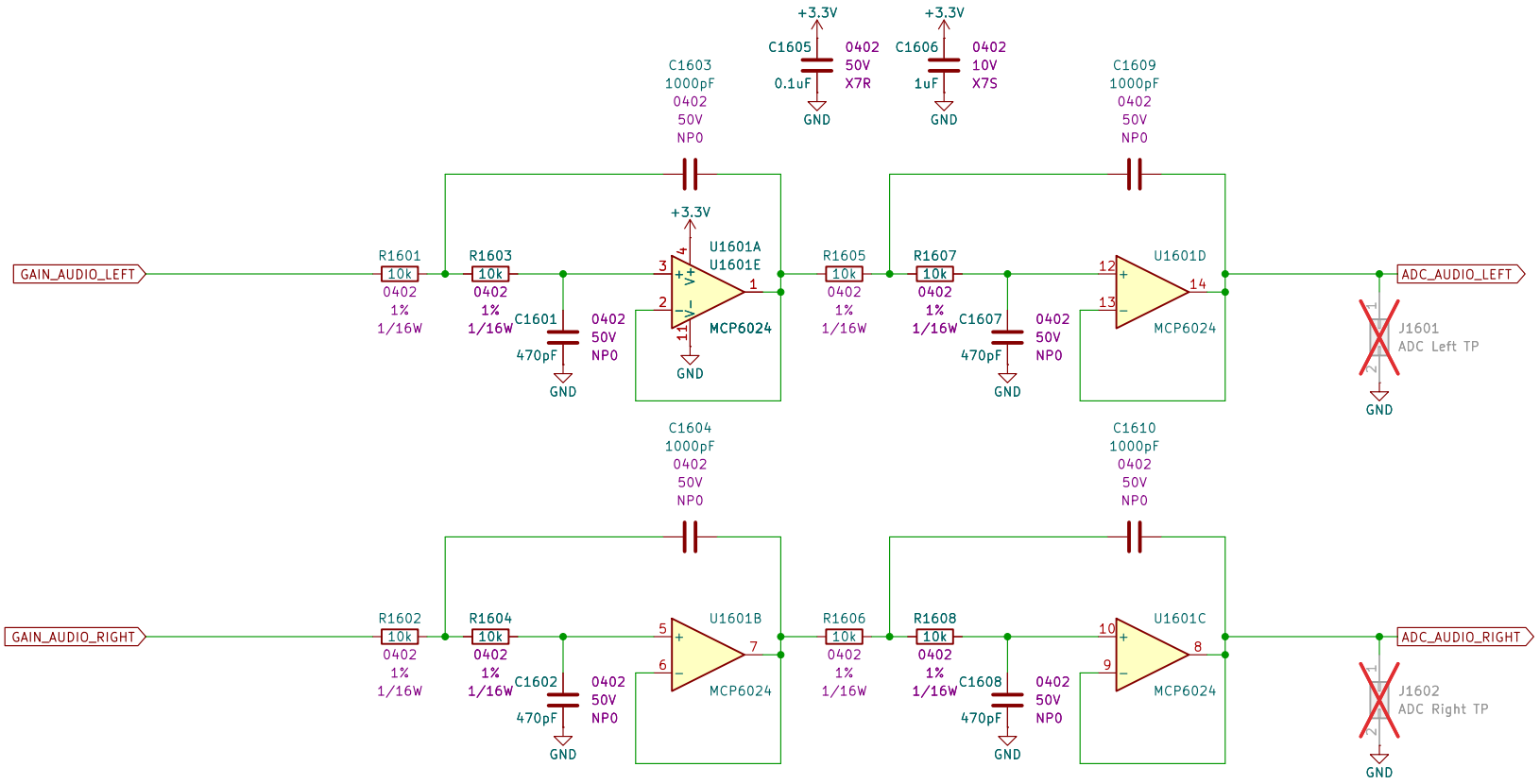


Drew Maatman and Drew Koetting		
Sheet: /Speaker Signal Conditioning/		
File: speaker_signal_conditoning.kicad_sch		
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Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 14/21



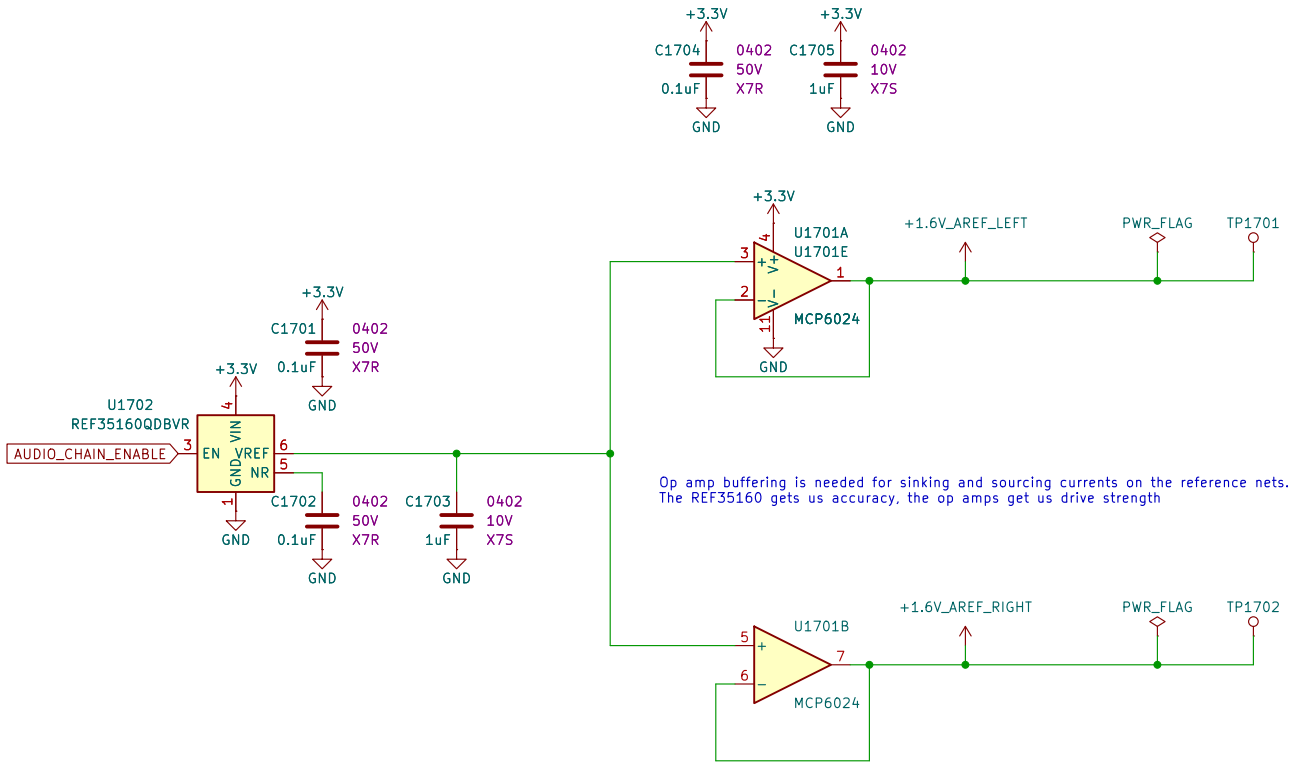
Drew Maatman and Drew Koetting	
Sheet: /Source Selection, PGAs/	
File: source_selection_pgas.kicad_sch	
Title: Audio Visualiser DSP	
Size: A	Date: 2026-10-27
KiCad E.D.A. 10.0.7	Id: 15/21

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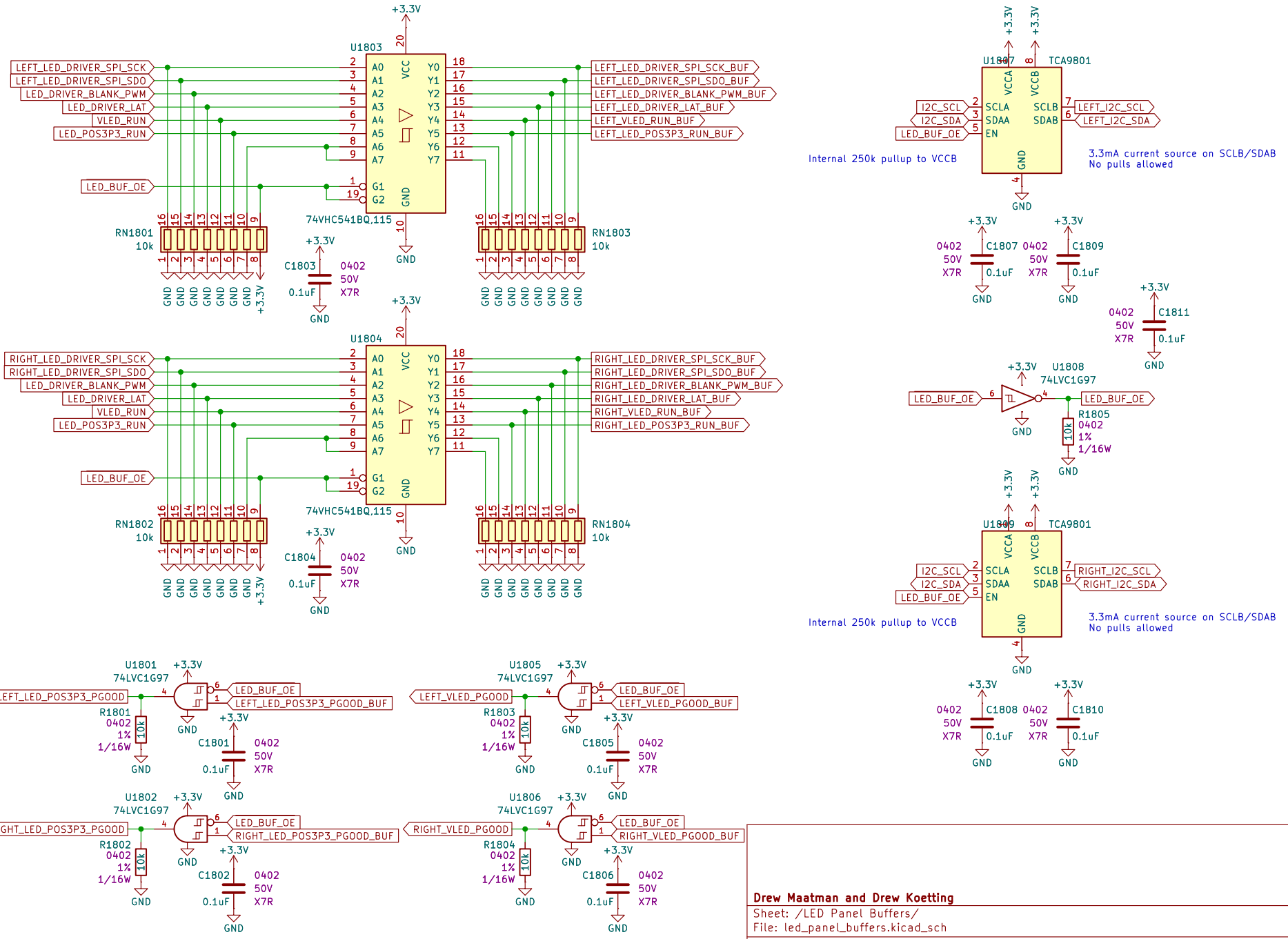


4-pole Sallen-Key active anti-aliasing lowpass filters
Unity gain
fc = 23.2kHz, Q = 0.707 each
If sampling at 44.1kHz, nyquist is 22.05kHz
Must filter hard around this frequency

Drew Maatman and Drew Koetting		
Sheet: /Anti-alias Filtering/		
File: anti_alias_filtering.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
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Drew Maatman and Drew Koetting		
Sheet: /Audio Voltage References/		
File: audio_voltage_references.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
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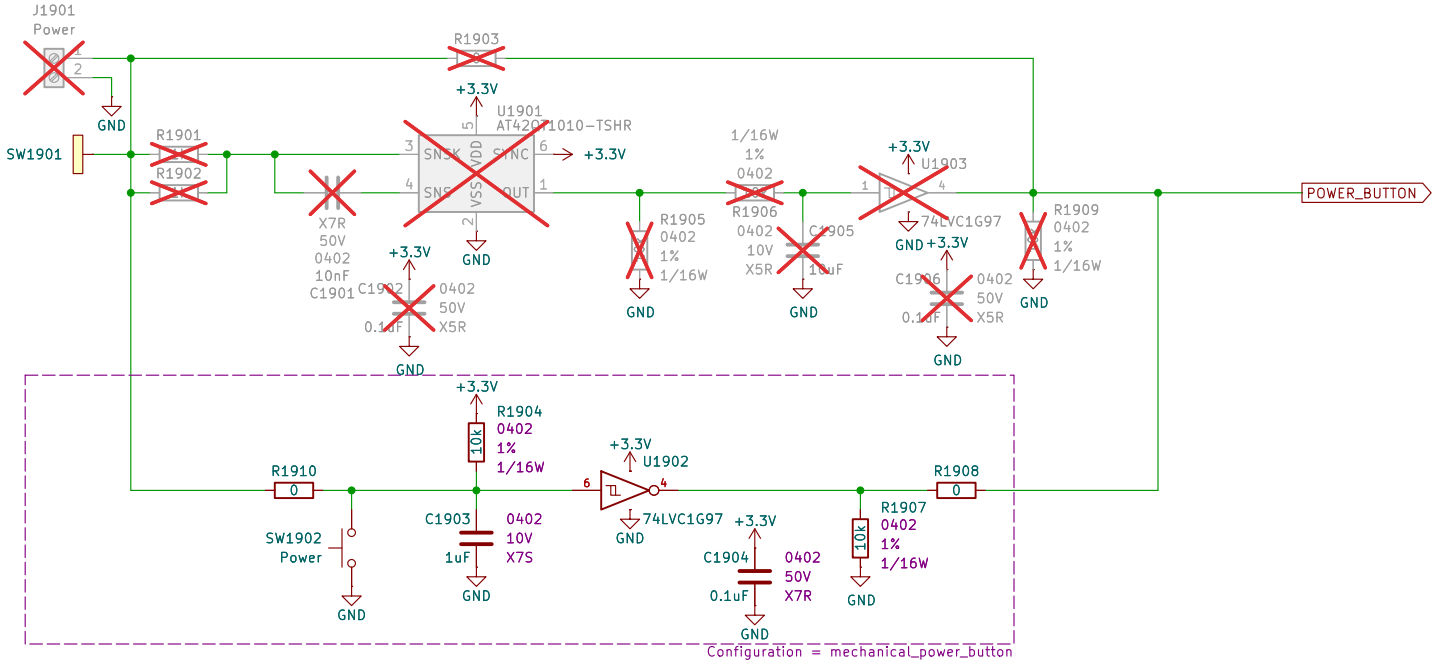
Drew Maatman and Drew Koetting

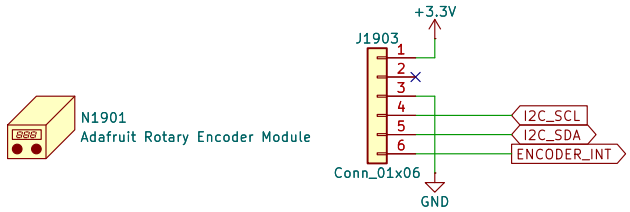
Sheet: /LED Panel Buffers/

File: led_panel_buffers.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
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I2C 7-Bit Address:
0x36

Drew Maatman and Drew Koetting

Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 19/21

H2001
MountingHole

H2003
MountingHole

H2002
MountingHole

H2004
MountingHole

Screw
MK2001

Screw
MK2005

Screw
MK2002

Screw
MK2006

Standoff
MK2003

Standoff
MK2007

Standoff
MK2004

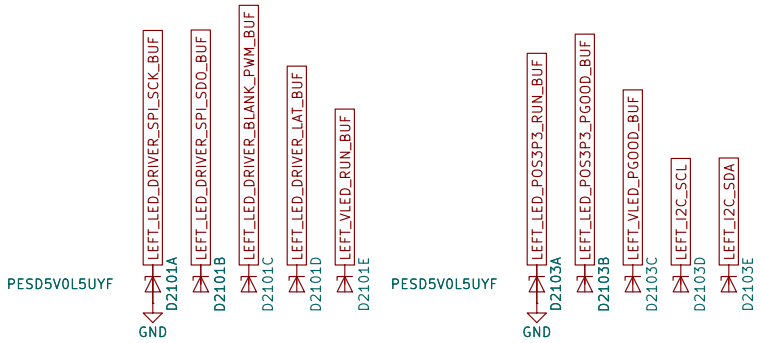
Standoff
MK2008

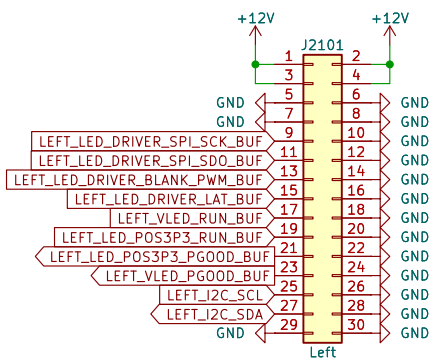
Drew Maatman and Drew Koetting

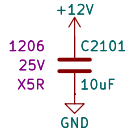
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File: mechanical.kicad_sch

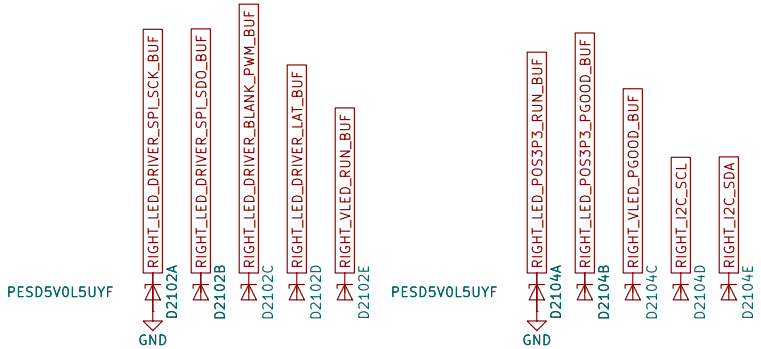
Title: Audio Visualiser DSP

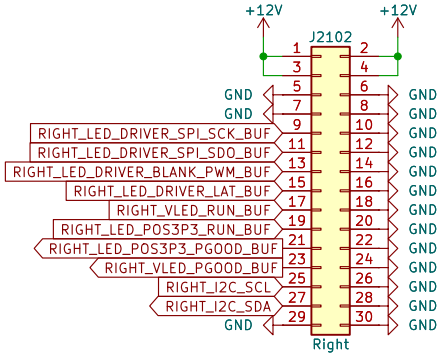
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 20/21

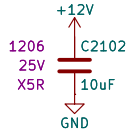












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Sheet: /LED Panel Connectors/		
File: led_panel_connectors.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 21/21